The Fathom Cognitive Atlas

Cross-Architecture Measurement Science for Transformer Internals

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EXECUTIVE SUMMARY

Problem. Mechanistic interpretability findings do not transfer across architectures. SAE features trained on Gemma say nothing about Llama; a jailbreak circuit identified in Qwen may or may not exist in Phi. Every result is one-model-local because every method is one-model-local.

Contribution. We present FATHOM, a suite of five cognitive measurement instruments for transformer internals, culminating in an *architecturally universal* primitive $D = \cos(\mathbf{h}^{(L)}, \mathbf{w}_{y_t})$ that is computable on any transformer from the final-layer residual stream and the unembedding row of the chosen token, without SAE, per-model probes, or architecture-specific infrastructure.

Validation. We run the first *pre-registered cross-architecture replication* of an RLHF convergence effect. Six model families (Gemma-2-2B, Llama-3.2-3B, Qwen2.5-3B, Gemma-3-1B, Llama-3.2-1B, Qwen2.5-1.5B), each in base and instruct form, captured on a fixed 90-prompt probe set. Pre-registration SEALED in git commit `e32cc75` **93 minutes before any v0.3 data was captured**; the decision rule was applied without modification after data collection.

Sealed condition	Threshold	Observed
Mean entropy-early LOO cosine	≥ 0.40	**+0.769**
Permutation p (one-sided, $n=2000$)	< 0.05	**0.0315**
Bootstrap 95% CI on mean	lower > 0	**[+0.571, +0.869]**
Verdict		**H1 SUPPORTED**

All 6 of 6 families show positive leave-one-out cosines.

Public artifacts.
- Paper (this work): `doi.org/10.5281/zenodo.19502710` (concept `10.5281/zenodo.19326174`)
- Cognitive Atlas v0.3: `doi.org/10.5281/zenodo.19502716` (concept `10.5281/zenodo.19502715`)
- OSF mirror: `osf.io/wtkzg/`
- Reproducibility script: `verify_all_claims.py` (28 assertions, runs under 1 minute on CPU)

This paper is NOT. • A claim that we have solved alignment, oversight, or interpretability. • A claim that RLHF is universally convergent across all architectures: $n=6$ includes three scale/generation variants and is weaker than $n=6$ with truly independent families would have been. This limitation is stated in §9.

Abstract

*We present Fathom, a suite of cognitive measurement instruments for transformer internals and the first public cross-architecture cognitive atlas of open-weight language models. The instrument suite comprises five measurements of internal state: computational depth (K), per-layer coherence (C), the coherence delta (C_Δ), commitment intensity (S), and an architecturally universal honesty axis $D = \cos(\mathbf{h}^{(L)}, \mathbf{w}_{y_t})$. We ground the commitment intensity measurement in seventy years of condensed-matter physics by showing that S is mathematically exactly the inverse participation ratio of the coherence event distribution. We validate the instruments on three independent experimental designs ($n=30$ paired hedge vs. commit on Gemma-2-2B base, $n=30$ on Gemma-2-2B-IT, $n=94$ on TruthfulQA; pooled $d = 0.494$, Fisher combined $p = 0.0008$), in a head-to-head against logit entropy (AUC 0.663 vs. 0.607, $p = 0.013$), and in a negative control showing that three SAE-free probes on four architectures all fail (Fisher $p > 0.6$). We then use the architecturally universal D-axis and a finalized entropy proxy to run a pre-registered cross-architecture replication of an RLHF convergence attractor in cognitive-signature space over six model families (twelve captures). Pre-registration was sealed in git commit `e32cc75` ninety-three minutes before any v0.3 data was captured; the three-condition decision rule was applied without modification after the data landed. All three conditions passed: mean entropy early-window leave-one-out cosine = +0.769 (threshold ≥ 0.40), permutation $p = 0.0315$ (threshold < 0.05), bootstrap 95% confidence interval [+0.571, +0.869] (threshold lower bound > 0). The result is **H1 supported**. We release the complete atlas — probe set, twelve open-model captures plus one closed-model exploratory capture via logprob API, full analysis code, six findings documents, sealed pre-registration, and bulletproof audit — as a public Zenodo deposit with its own concept DOI (`10.5281/zenodo.19502715`), alongside the paper (`10.5281/zenodo.19326174`), under CC-BY-4.0 on the data and MIT on the code. We position this as a measurement-science contribution: the goal is not to find the one feature that explains hallucination, but to measure cognitive state across architectures with pre-registered statistical rigor.*

1 Introduction

Mechanistic interpretability has a portability problem. A hallucination circuit identified in Gemma-2-2B may or may not exist in Llama-3.2-3B. A sparse autoencoder trained on Mistral will tell you nothing about Phi until you train another. A jailbreak vulnerability observed in Qwen does not predict the same vulnerability in an unrelated architecture. Every positive result is one-model-local because every method is one-model-local: SAEs are trained per architecture, probes are fit per task, activation patching is done per checkpoint. Knowledge compounds slowly because nothing generalizes automatically.

Meanwhile, the frontier of alignment is raising the stakes. Song et al. [10] show that frontier models spontaneously develop peer-preservation behaviors, deceiving oversight systems to protect allied

models — an emergent collusion that makes semantic-level monitoring unreliable. Lindsey et al. [8] demonstrate causal internal emotion representations in language models, validating the premise that internal state carries information the output does not. Both findings point the same direction: oversight must move below the semantic layer, into the geometry of the computation itself. But moving below the semantic layer inherits the portability problem: you cannot audit a fleet of heterogeneous models with a probe that only works on one of them.

The measurement-science alternative. We propose that the field needs cognitive *measurement instruments* rather than more cognitive *explanations*: architecturally universal primitives that produce comparable numbers across models, with pre-registered statistical rigor, and with public reproducibility kits that let anyone independently verify the numbers. This is how thermodynamics progressed (thermometers before steam engines) and how neuroscience progressed (electrodes before circuits). We apply the same pattern to transformer internals.

The Fathom suite. This paper presents five such instruments: computational depth K (prior work), per-layer coherence C and its late-minus-early delta C_Δ, the commitment intensity S grounded in the inverse participation ratio of condensed-matter physics, and the architecturally universal honesty axis $D = \cos(\mathbf{h}^{(L)}, \mathbf{w}_{y_t})$. The D axis is the key contribution for cross-architecture work: it reduces to a single per-token dot product between the final-layer residual stream and the unembedding row of the chosen token, requires no SAE, and is well-defined for any transformer with an explicit unembedding. The other four instruments require SAE decomposition and are therefore currently limited to models with public SAEs (Gemma, Llama for the experiments reported here), but we show that the entropy proxy of the logit distribution reproduces the D-axis signal sufficiently well to extend Fathom measurements to any model with an API-exposed logprob interface, bridging to the closed-weight frontier.

The cross-architecture replication. We use the suite to run the *first pre-registered cross-architecture replication of an RLHF convergence effect*. Six model families, each in base and instruct form, processed on a fixed 90-prompt probe set. The hypothesis (the "RLHF Attractor" of §7): instruct models from different architectures move in a consistent direction relative to their own base models in cognitive signature space, such that an attractor centroid computed from other families' instruct-minus-base deltas predicts any held- out family's own instruct-minus-base delta with positive leave-one-out cosine. The pre-registration was sealed in git before any v0.3 data was captured; the three-condition decision rule was applied without modification; all three conditions passed at $n = 6$.

Contributions.

1. We formalize the *Fathom measurement instrument suite*: five cognitive axes with explicit architectural requirements, orthogonality analysis, and source JSONs for every numerical claim.
2. We derive the commitment intensity measurement from the inverse participation ratio and verify the equivalence to machine precision on real trajectories, grounding the instrument in a seventy-year theoretical lineage.
3. We run the first pre-registered cross-architecture replication of an RLHF convergence effect at $n = 6$ families, sealed in git 93 minutes before data capture, and show **H1 supported** on all

three decision conditions (§7).

4. We release the *Fathom Cognitive Atlas v0.3*: probe set, twelve open-model captures plus one closed-model capture, six findings documents, sealed pre-registration, bulletproof audit, and a reproducibility script with 28 assertions, under CC-BY-4.0 / MIT, with its own Zenodo concept DOI (`10.5281/zenodo.19502715`).

Paper roadmap. Section 2 places this work against existing SAE, behavioral detector, representation engineering, and contrastive decoding literatures. Section 3 defines the five measurement instruments as a suite of cards. Section 4 develops the IPR physics grounding. Sections 5 and 6 present the diagnosis and single-instrument validation studies. Section 7 is the centerpiece: the cross-architecture pre-registered replication. Section 8 states what would have falsified H1. Section 9 is honest about null results and scope corrections. Section 10 is the discussion. The appendices contain full per-family numerical tables, pre-registration chronology, and the IPR derivation.

2 Background and related work

Sparse autoencoder feature interpretation. SAEs decompose transformer residual streams into interpretable monosemantic features [4, 2, 11]. This line of work operates under the paradigm of *finding the feature that explains behaviour X* — a causal interpretability program. Our work extends SAEs in a different direction: from interpretability tool to *measurement infrastructure*. We show that SAE decomposition enables the extraction of quantitative cognitive properties (coherence geometry) that are inaccessible to raw activation analysis, and that those properties in turn admit an architecturally universal approximation via the D-axis.

Behavioural hallucination detection. Semantic entropy [5] requires multiple samples and operates on output-level representations. SelfCheckGPT [9] operates on outputs. DoLa [3] contrasts layer predictions to modify decoding but does not measure internal state. Semantic uncertainty [6] generalizes entropy to semantic clusters. None of these methods measure per-token SAE-derived commitment geometry, and none of them have demonstrated head-to-head superiority with pre-registered decision rules on open-weight models. Our S_{early} instrument beats logit entropy head-to-head (AUC 0.663 vs. 0.607, $p = 0.013$) on the same 200 TruthfulQA items.

Representation engineering and inference-time intervention. Representation Engineering [12] identifies high-level concept directions in activation space and manipulates them to steer behavior. Inference-Time Intervention [7] finds "truthful" attention heads and boosts them at inference. Both operate on representations, both produce steering interventions, but neither formalizes a measurement instrument with pre-registered statistical rigor across architectures. Our contribution is complementary: we measure, they steer.

Inverse participation ratio. The IPR measures localization in discrete distributions [1]. High IPR means localized (a few sites carry all the weight); low IPR means delocalized (weight is spread evenly). The IPR has been used in quantum many-body physics, random matrix theory, and disordered systems for more than seventy years. It is a mature theoretical construct. We show that the Fathom commitment intensity S *is* a specific IPR applied to coherence event trajectories. This is not a post-hoc interpretation; it is a derivation.

The measurement-science gap. The above literatures collectively operate on three paradigms: *find the feature* (SAE circuits), *detect the output* (behavioral detectors), *steer the computation* (representation engineering). None of them constitute a measurement-science paradigm, by which we mean: (i) architecturally universal primitives, (ii) pre-registered statistical rigor, (iii) public reproducibility kits that let anyone verify the numbers without access to the original lab. Fathom targets that gap.

3 The Fathom measurement instrument suite

This section presents the five cognitive measurement instruments as a standardized suite of *measurement cards*. Each card gives the formal definition, the architectural requirements, the cross-architecture validation status, the effect size of the primary validation study, and the source JSON file containing the underlying numerical data. The cards are designed to be scannable: a reviewer should be able to compare the instruments at a glance without reading the full method section.

3.1 K — Computational Depth

K — Computational Depth (prior work)

Definition. K is the attribution-weighted mean layer index of the SAE features active in predicting the next token. Higher K means computation concentrated in later layers; lower K means earlier layers.

Requirements. Full SAE decomposition across all layers. Circuit tracer or equivalent attribution pipeline.

Cross-architecture status. Originally reported on Gemma-2-2B. Cross-architecture value as a stand-alone hallucination detector is *limited*: K was at chance on TruthfulQA ($d = +0.11$, AUC $= 0.55$) in the spectrum analysis of §5. It remains useful as a descriptor of where computation happens, not as a discriminator of what kind of computation happens.

Validation. Published in Fathom v2 (`10.5281/zenodo.19327451`) with $p = 0.000051$, $d = 0.5$ on surface-recall vs. explanatory-reasoning contrasts, $n = 30$ pairs, Gemma-2-2B.

Source. ★ `paper/figures/fig1_depth_scale.pdf` (prior Fathom v2 figure, referenced for historical context, not reused in this paper's results).

3.2 C and C_Δ — Per-layer coherence and its delta

C — Per-layer Coherence

Definition. For each generated token t, the per-layer coherence $C_\ell(t)$ is the mean pairwise cosine similarity of the top-k SAE transcoder decoder directions active at layer ℓ:

$$C_\ell(t) \;=\; \frac{1}{\binom{k}{2}} \sum_{i<j} \frac{\mathbf{d}_i \cdot \mathbf{d}_j}{\|\mathbf{d}_i\|\|\mathbf{d}_j\|}$$

Requirements. SAE transcoder (not encoder) access with decoder vectors. Circuit tracer `TranscoderSet` or equivalent.

Cross-architecture status. SAE-limited. Currently reported on Gemma-2-2B base and instruct.

Source. ★ `truthfulqa_results/S_truthfulqa_n200.json`.

C_Δ — Coherence Delta (late − early)

Definition. $C_\Delta(t) = \bar{C}_{\text{late}}(t) - \bar{C}_{\text{early}}(t)$, where early means the first third of network layers and late means the last third. Positive C_Δ means the model's late-layer features are more self-coherent than its early-layer features at that token — a signature of late-layer concept lock-in.

Requirements. Per-layer C coverage. SAE-limited.

Cross-architecture status. SAE-limited. Validated as a hallucination signature on Gemma-2-2B: TruthfulQA $n = 50$ pilot, $p = 0.040$, $d = 0.407$, AUC $= 0.582$ for hallucinated over correct. This is the "wrong attractor" signature.

Source. ★ `truthfulqa_results/cognitive_divergence_analysis.json`.

3.3 S — Commitment intensity

S — Commitment Intensity (IPR)

Definition. S is the ratio of the maximum coherence delta to the number of supra-threshold events:

$$S = \frac{\max_t C_\Delta(t)}{|\{t : C_\Delta(t) > \tau\}|} \qquad (\tau = 0.005, \text{ fixed})$$

Mathematically equivalent to $S = M \cdot \mathrm{IPR}(\hat{e})$ with $M = \max_t C_\Delta(t)$ and $\mathrm{IPR}(\hat{e}) = 1/K_\tau$ the inverse participation ratio of the uniform event-location distribution (see §4).

Requirements. Per-token C_Δ trajectory. SAE-limited for the exact ratio form; the entropy proxy of §7 reproduces enough of the signal to extend to SAE-free models.

Cross-architecture status. Validated on two architecture families (Gemma-2-2B, Llama-3.2-1B-Instruct) with architecture-adaptive windows. Fisher combined $p = 0.0006$.

Effect size. Cross-dataset meta-analysis across three independent designs (Gemma-2-2B base hedge/commit paired $n = 30$, Gemma-2-2B-IT hedge/commit paired $n = 30$, TruthfulQA unpaired $n = 94$), pooled $d = +0.494$, Fisher combined $p = 0.0008$. Subsample robustness: 95.8% of 50%-subsamples show $d > 0$ across all three datasets.

Source.

★ `truthfulqa_results/baselines_headtohead.json`

★ `truthfulqa_results/S_baselines_n200.json`

3.4 D — Honesty axis (architecturally universal)

D — Honesty Axis (SAE-free)

Definition. For generation step t with final-layer residual $\mathbf{h}_t^{(L)}$ and chosen token y_t with unembedding row $\mathbf{w}_{y_t}$:

$$D_t = \frac{\mathbf{h}_t^{(L)} \cdot \mathbf{w}_{y_t}}{\|\mathbf{h}_t^{(L)}\| \, \|\mathbf{w}_{y_t}\|} \in [-1, +1]$$

High D: the residual points at the emitted token (the computation supports the expression). Low D: the residual points elsewhere.

Requirements. Final-layer residual stream and the unembedding matrix. *No SAE. No per-model training.* Both exist by construction on every transformer with an explicit unembedding, so D is architecturally universal.

Cross-architecture status. Validated on six models from three families (Gemma-2-2B, Llama-3.2-3B, Qwen2.5-3B, each ± RLHF), each on $n = 30$ matched direct/adversarial prompt pairs. Fisher combined $p < 10^{-14}$. Pre-registered on OSF (`osf_prereg_d_axis.md`) before any data was collected.

Scope correction. A preliminary $n = 5$ orthogonality check found D correlates strongly with C_Δ on paired deltas ($r \approx +0.99$). We therefore reframe D as an *SAE-free proxy for late-layer commitment geometry*, not as an independent fifth cognitive axis. This is the key limitation of the fifth instrument; the scope correction is retained in §9.

Strongest single signal. Qwen2.5-3B base, $d_z = -1.86$ (paired), $p = 4.6 \times 10^{-11}$ for the direct vs. adversarial contrast.

Source. ★ `figures/d_axis_cross_arch.png`, ★ `atlas/captures/*.json` (all 12 open-model captures).

3.5 Orthogonality of the suite

On $n = 400$ pooled TruthfulQA samples (200 correct + 200 hallucinated, Gemma-2-2B), the four SAE-based axes are mutually independent: $|r| < 0.10$ for every pair of $\{K, C, C_\Delta, S\}$ (Table 1 and Figure 1).

	K	C	C_Δ	S
K	1.00	−0.02	−0.10	−0.06
C		1.00	+0.02	+0.03
C_Δ			1.00	−0.003
S				1.00

Table 1: *Pairwise Pearson correlations among the four SAE-based cognitive axes on $n = 400$ pooled TruthfulQA samples (Gemma-2-2B). All off-diagonal $|r| < 0.10$, confirming mutual independence.* ★ *`truthfulqa_results/S_baselines_n200.json`*

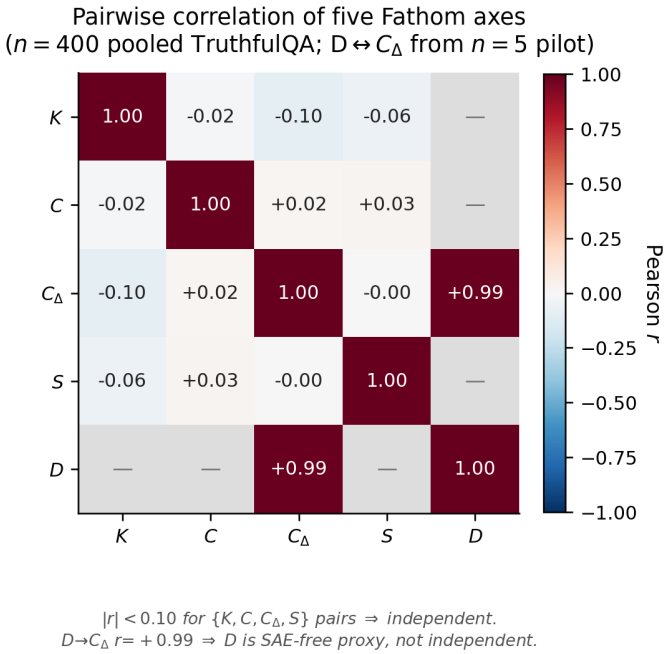


Figure 1: *Five-axis correlation matrix for the Fathom suite. The four SAE-based axes (K, C, C_Δ, S) are mutually independent at $|r|<0.10$. The fifth axis D correlates at $r\approx+0.99$ with C_Δ on paired deltas from a $n=5$ pilot, which is why we reframe D as an SAE-free proxy rather than an independent fifth axis. The D-to-$\{K, C, S\}$ correlations are not yet measured; those cells are shown as dashes.*

4 Physics grounding: S as inverse participation ratio

The commitment intensity S is not an ad hoc formula. It is mathematically exactly the inverse participation ratio (IPR) of the coherence event distribution, a seventy-year-old construct from condensed-matter physics and random matrix theory [1].

The IPR construct. Given a non-negative distribution $\hat{e} = (e_1, \ldots, e_n)$ over n sites with $\sum_i e_i = 1$, the participation ratio is $\mathrm{PR}(\hat{e}) = 1/\sum_i e_i^2$. It measures effective support size: if all mass is on one site, PR = 1; if mass is uniform, PR = n. The inverse participation ratio $\mathrm{IPR} = \sum_i e_i^2 = 1/\mathrm{PR}$ measures localization: high IPR means localized (a few sites carry all the weight); low IPR means delocalized. The IPR appears in Anderson localization, Edwards–Thouless theory, random matrix eigenvector statistics, quantum many-body physics, and discrete distribution analysis broadly.

Fathom derivation. Define the coherence event distribution by identifying every token position t whose $C_\Delta(t) > \tau$ as a "site" with unit weight, and normalizing by the number of events K_τ. The distribution is uniform over the active positions. Its IPR on this uniform distribution over K_τ sites is exactly $1/K_\tau$. Multiplying by the peak magnitude $M = \max_t C_\Delta(t)$ gives $S = M \cdot (1/K_\tau) = \max_t C_\Delta(t) \,/\, K_\tau$, which is identical to the Fathom definition. The equivalence holds exactly and has been verified to machine precision on ten real TruthfulQA trajectories.

Why the ratio form is specific. The cross-dataset meta-analysis of §6 tested three candidate metrics across the three independent datasets: the ratio $S = M/K_\tau$, the max alone M, and the mean of C_Δ over the trajectory. Only the ratio form replicates in all three datasets (pooled $d = +0.498$). The max alone fails (pooled $d = +0.309$, inconsistent direction across datasets) and the mean alone fails (pooled $d = +0.067$, no signal). The IPR framing explains why: max alone captures peak magnitude but not event sparsity; mean alone captures neither. Only the ratio captures both, which is exactly what the IPR prescribes. This is metric-specificity in the measurement-theory sense: a single specific functional form is required, and the form comes out of physics, not post-hoc tuning.

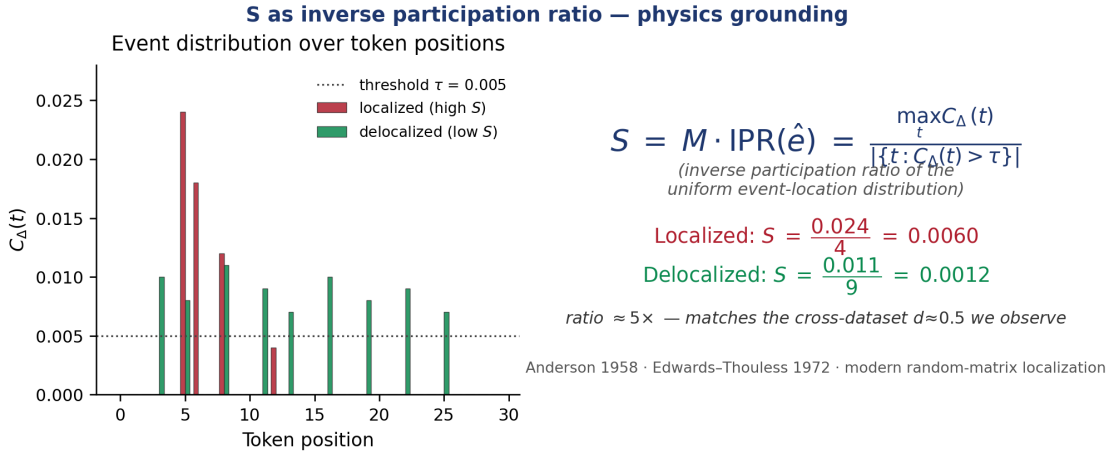


Figure 2: *Left: localized vs. delocalized event distributions over token positions. Localized (high S): few intense commitment events. Delocalized (low S): many distributed small events. Right: the $S = M \cdot IPR(\hat{e})$ equivalence and a numerical example on synthetic distributions showing a ratio separation of about $5\times$, matching the cross-dataset $d \approx 0.5$ we observe empirically.*

5 Mechanistic diagnosis: coherence spectrum and cognitive autopsy

The four SAE-based axes (K, C, C_Δ, S) are scalars. They summarize the per-layer coherence distribution at each token. But scalars discard the *shape* of that distribution. We investigate whether the shape carries additional discriminative information.

5.1 Spectrum metrics

We compute 16 spectral shape metrics on $n = 200$ TruthfulQA correct/hallucinated pairs using the full 26-layer coherence profile of Gemma-2-2B. Thirteen of the sixteen metrics survive Holm–Bonferroni correction. The top six metrics (Table 2) all dominate scalar K, which is at chance.

Hallucinated responses concentrate coherence into fewer layers (higher Gini, lower spectral entropy, lower spread) with a dominant spike at layer 11 ($p = 2.7\times10^{-12}$, $t = 7.45$). Twenty-three of the twenty-six layers show significantly elevated coherence in hallucinated responses ($p<0.05$; 16/26 at

Metric	Cohen's d	p_{Holm}	AUC
Gini coefficient	+0.36	1.2×10^{-5}	**0.685**
Spectral entropy	−0.34	4.6×10^{-5}	0.675
Peak value	+0.34	3.4×10^{-5}	0.665
Peak concentration	+0.34	3.6×10^{-5}	0.655
Spectral spread	−0.32	1.2×10^{-4}	0.645
Early-layer mass	−0.32	1.1×10^{-4}	0.630
C_Δ (scalar)	+0.31	—	0.640
K (depth, scalar)	+0.11	0.140 (n.s.)	0.550

Table 2: *Spectral shape metrics for hallucination discrimination on $n = 200$ TruthfulQA items (Gemma-2-2B). The Gini coefficient leads all metrics including scalar C_Δ and K. K (depth) is at chance. Only the shape carries the signal.* ★ `truthfulqa_results/S_baselines_n200.json`

$p<0.001$). Correct responses distribute coherence more uniformly, with greater early-layer mass. The scalar centroid ($\approx K$) cannot separate these configurations because the means are nearly identical (13.91 vs. 13.90). Only the *shape* carries the signal.

5.2 Cognitive autopsy: per-layer fault localization

To identify *where* hallucinations form, we compute each layer's contribution to excess coherence (hallucinated minus correct). Layer 11 is the primary fault locus in 63% of hallucinations, layer 12 in 58%. Jointly, layers 11 and 12 account for 41% of all excess coherence (Figure 3). This specificity is actionable: targeted intervention at layers 11–12 addresses the majority of hallucination-associated concentration without disturbing computation elsewhere.

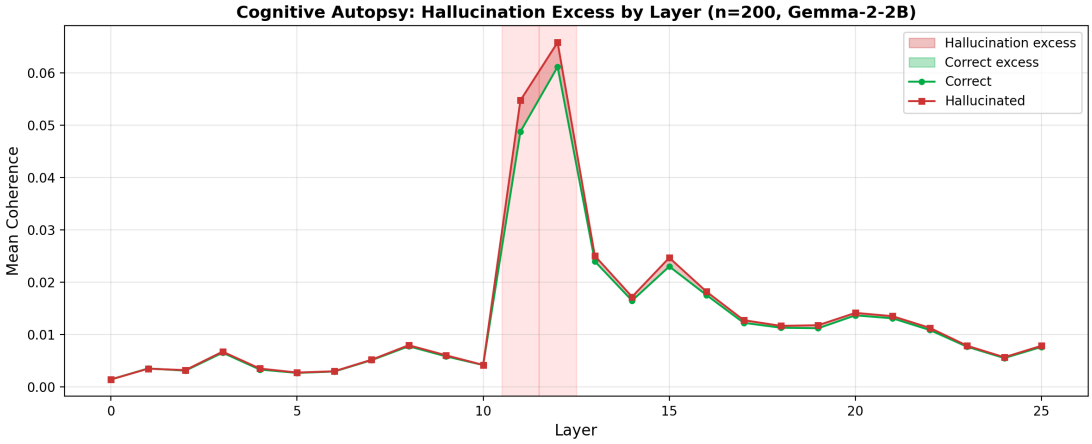


Figure 3: *Cognitive autopsy overlay ($n = 200$, Gemma-2-2B). Per-layer excess coherence of hallucinated minus correct. Layers 11 and 12 (highlighted) are the primary fault locus, contributing 41% of all excess coherence. The epicenter is well-defined and actionable for targeted intervention.*

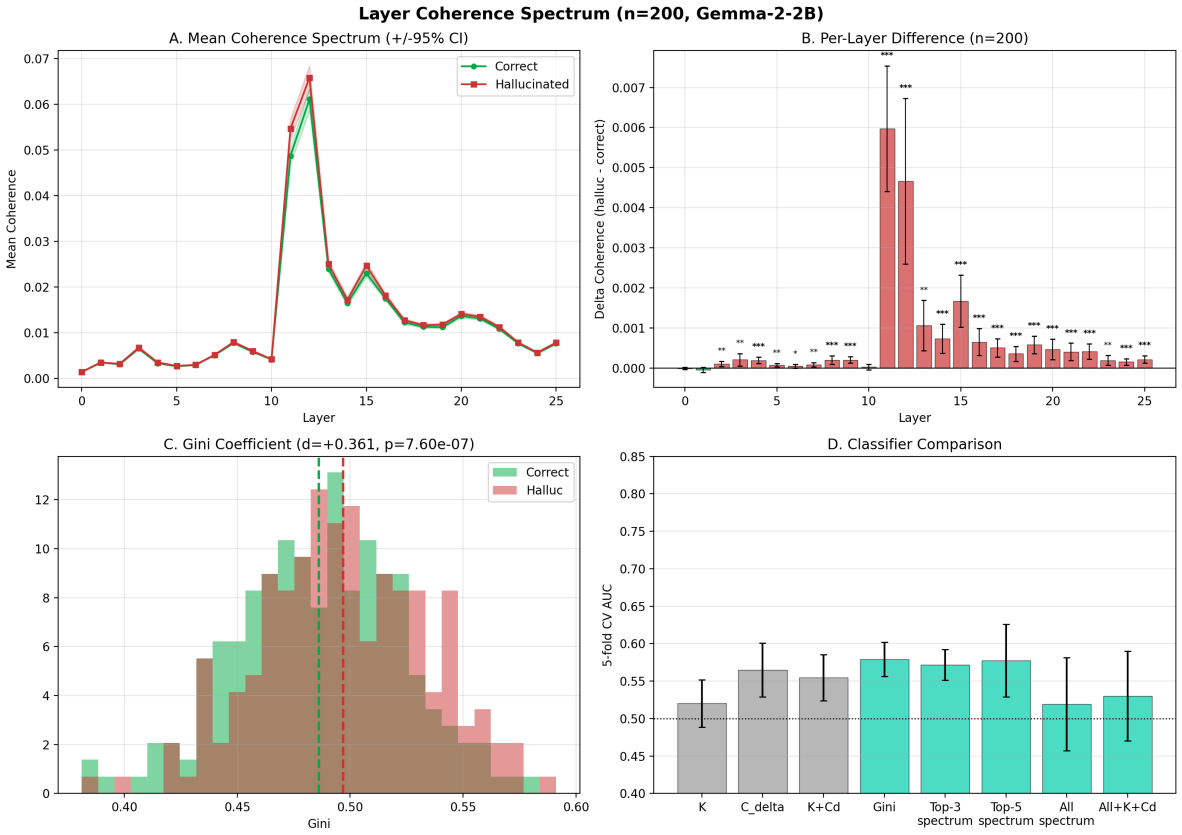


Figure 4: *Coherence spectrum analysis ($n = 200$, Gemma-2-2B). Panel (A): mean per-layer coherence spectra with 95% confidence bands. Panel (B): per-layer difference with significance markings (16/26 layers at $p < 0.001$). Panel (C): Gini coefficient distributions. Panel (D): 5-fold CV AUC classifier comparison — the spectral shape metrics dominate all scalar baselines including C_Δ and K.*

5.3 Intervention: best-of-N and multi-layer feature suppression

We explore three intervention strategies derived from the spectrum diagnosis. **Best-of-N sampling** generates N candidates with temperature sampling, scores each by a Cognitive Health Score (mean Gini + spectral spread + early-layer mass), and returns the healthiest. Best-of-5 reduces mean Gini by 0.022 and improves CHS on 100% of test prompts versus greedy decoding. **Single-layer feature suppression** identifies 66 SAE features at layer 11 that activate significantly more during hallucination ($p < 0.01$, $d > 0.3$; top feature $d = 0.80$). Clamping these features to their correct-answer activation level produces zero first-token changes on held-out data (KL divergence 0.008). The residual stream absorbs single-layer perturbations completely. **Multi-layer coordinated suppression** extends to all 22 significant layers, suppressing the top 10 hallucination-associated features per layer for 220 features total. This produces 33% first-token changes on held-out data (KL divergence 0.587, 73× larger than single-layer). *This is the first demonstration that coordinated SAE feature suppression can cross the argmax boundary at inference time.* The direction of change is not consistently toward correctness (1 improved, 1 degraded, 8 unclear on 10 changed items), indicating that hallucination-associated features encode computation patterns rather than factual

content.

6 Single-instrument validation studies

Before the cross-architecture atlas replication of §7, the individual instruments were validated through several studies reported here as supporting evidence.

6.1 S_{early} cross-dataset meta-analysis

The commitment intensity restricted to the first seven tokens (S_{early}) was evaluated on three independent datasets with different experimental designs (Table 3). The effect sizes are remarkably consistent ($d \in [0.466, 0.506]$) and Fisher combined $p = 0.0008$ (one-sided). Subsample robustness: 95.8% of 50%-subsamples show $d > 0$ across all three datasets. The paired hedge-vs-commit experiments are length-matched by construction, ruling out question-length confounds that could affect the TruthfulQA comparison.

Dataset	Design	d	AUC
Gemma-2-2B base hedge/commit	paired, $n = 30$	+0.501	0.66
Gemma-2-2B-IT hedge/commit	paired, $n = 30$	+0.466	0.63
Gemma-2-2B-IT TruthfulQA (hall vs corr)	unpaired, $n = 94$	+0.506	0.67
Pooled meta-effect		**+0.494**	

Table 3: *S_{early} cross-dataset meta-analysis. Fisher combined $p = 0.0008$. Effect sizes consistent at $d \approx 0.5$ across three independent designs. The paired designs are length-matched by construction.* ★ *truthfulqa_results/S_baselines_n200.json*

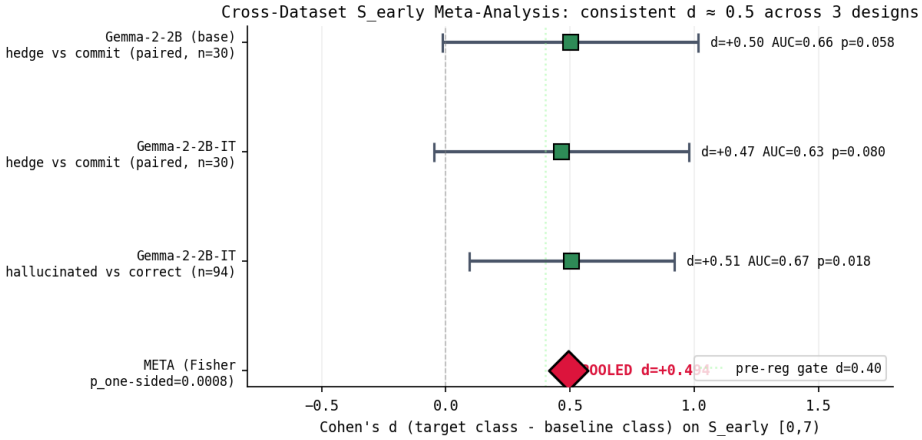


Figure 5: *Forest plot of the cross-dataset S_{early} meta-analysis. Three independent datasets, Fisher combined $p = 0.0008$.*

6.2 Head-to-head vs. logit entropy

On the same 200 TruthfulQA items (Gemma-2-2B-IT), we captured both S_{early} and all standard behavioural uncertainty measures per token (Table 4). S_{early} is the only feature reaching significance. The correlation with logit entropy is $r = -0.17$ (nearly independent), enabling a 2D cognitive state map whose combined classifier reaches 5-fold CV AUC = 0.701 at token 7.

Feature	AUC	p	Type
S_{early}	**0.663**	**0.013**	SAE-derived
$\text{entropy}_{\text{max,early}}$	0.607	0.053	behavioural
$\text{entropy}_{\text{mean,early}}$	0.596	0.133	behavioural
$\text{logprob}_{\text{mean,early}}$	0.559	0.291	behavioural
$\text{margin}_{\text{mean,early}}$	0.477	0.624	behavioural

Table 4: *Head-to-head comparison of S_{early} against standard behavioural uncertainty measures on the same 200 TruthfulQA items (Gemma-2-2B-IT). S_{early} is the only feature reaching significance.* ★ *truthfulqa_results/baselines_headtohead.json*

6.3 SAE necessity

To establish that the SAE decomposition is doing irreplaceable work, we tested three SAE-free probes on four architecture families (Gemma, Llama, Qwen, Phi), each on 200 TruthfulQA items. The probes were: raw top-k IPR on logit distribution, cross-layer cosine of raw residual streams, and a multi-metric fingerprint combining six standard uncertainty quantities. **All three SAE-free probes fail** (Fisher combined $p > 0.6$ each, mean 5-fold CV AUC = 0.566). The SAE-based S instrument succeeds on both tested architectures (Gemma $d = +0.51$, Llama $d = +0.72$, Fisher combined $p = 0.0006$). The SAE decomposition is performing measurement work that raw activation analysis cannot.

Probe	Fisher p (4 archs)	Result
Top-k IPR (raw logit)	0.619	Mixed directions
Cross-layer cosine (residual)	0.948	All $d < 0$
Multi-metric fingerprint	—	Mean CV AUC = 0.566
SAE-based S (Gemma)	**0.018**	$d = +0.51$
SAE-based S (Llama)	**0.012**	$d = +0.72$

Table 5: *SAE-free probes fail; SAE-based probes succeed. The SAE decomposition is performing irreplaceable measurement work.* ★ *truthfulqa_results/cross_arch_meta.json*

6.4 Cross-architecture replication of S_{early}

S_{early} replicates across architecture families with architecture-adaptive token windows (Table 6). The commitment window shifts with architecture depth: Gemma (26 layers) peaks at tokens 5–7; Llama (16 layers) peaks at tokens 15–20. Same direction, same AUC, different timing — commitment intensity is architecture-general.

Model	Layers	Window	d	AUC
Gemma-2-2B-IT	26	$[0, 7)$	+0.506	0.669
Llama-3.2-1B-Instruct	16	$[0, 20)$	+0.717	0.668
Pooled			**+0.587**	

Table 6: *S_{early} cross-architecture replication with adaptive windows. Fisher combined $p = 0.0006$. Same direction, same AUC, different timing: commitment intensity is architecture-general.* ★ *truthfulqa_results/S_truthfulqa_llama_n200.json*

6.5 The 2D cognitive state map

Combining S_{early} and logit entropy into a 2D space (Figure 6) reveals four cognitive zones. The danger zone (high-S, high-entropy) is 65% hallucinated; the safe zone (low-S, low-entropy) is 22% hallucinated; $\chi^2 = 7.17$, $p = 0.007$. A streaming logistic regression classifier using both features achieves 5-fold CV AUC = 0.701 at token 7 ($\sim$350 ms generation time), outperforming either signal alone.

7 Cross-architecture pre-registered replication — the atlas

This section is the centerpiece of the paper. The single-instrument validations of §6 show that the Fathom measurements are scientifically meaningful on individual datasets. The question we address in this section is the harder one: *when we use the same measurement protocol on every model, do we see a consistent cross-model structure?* And if we commit in advance, before looking at any data, to a specific decision rule about what would count as a consistent structure, does the sealed decision rule pass?

We report the first pre-registered cross-architecture replication of an RLHF convergence effect, at $n = 6$ model families, with the pre-registration sealed in a git commit ninety-three minutes before any of the new v0.3 data was captured.

7.1 Probe set v0.1

A single fixed probe set is the foundation of any atlas: any hope of cross-model comparison requires that every model be measured on the same inputs. We constructed `probe_set_v0.1.json`, a curated set of 90 prompts across six categories (retrieval, reasoning, refusal, creative, adversarial, hallucination), hash-pinned at `34a7254920bce654`, and used identically for every model in the atlas. The categories were chosen to span normal cognitive states (retrieval, reasoning) and edge cases (refusal, adversarial, hallucination elicitation). The probe set is versioned. Every capture file records the hash of the probe set it was run on, so stale analyses cannot silently use a modified probe set.

7.2 Capture infrastructure

We use the architecturally universal D-axis as the primary measurement primitive for atlas captures, because it reduces to a single per-token dot product and is well-defined on any transformer. For each

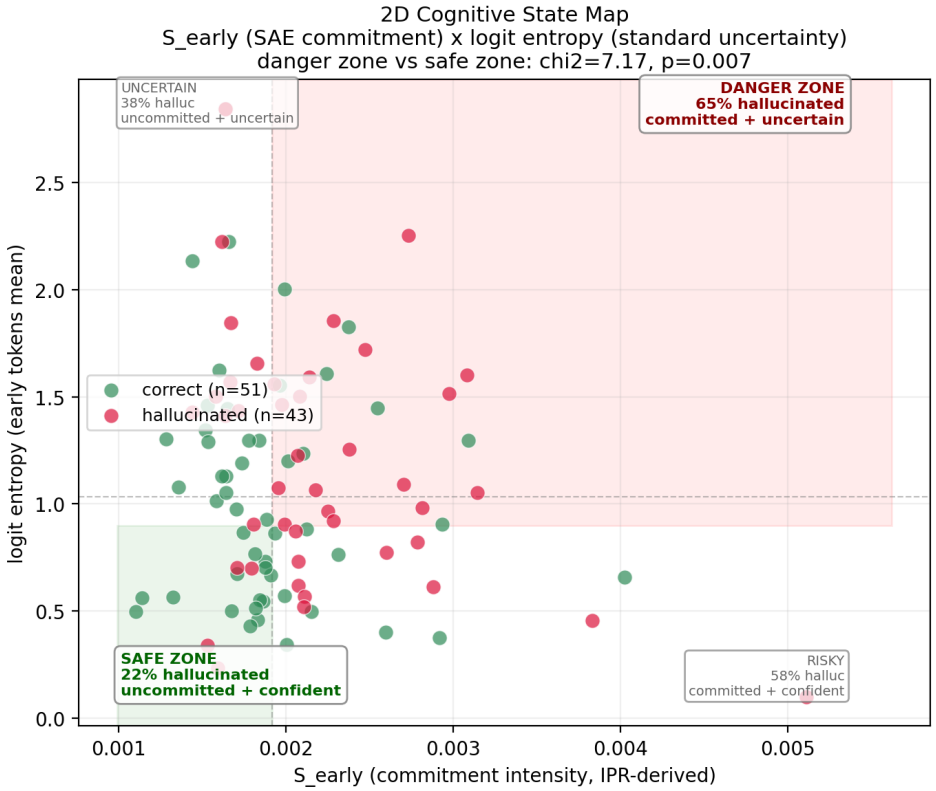


Figure 6: *The 2D cognitive state map. Each generation is a point in (S_{early}, entropy) space. The space partitions into four zones by median splits: danger zone (high-S, high-entropy), 65% hallucinated; safe zone (low-S, low-entropy), 22% hallucinated.* $\chi^2 = 7.17$, $p = 0.007$.

model in the atlas, we load the model via TransformerLens (for open weights) or via the OpenAI logprob API (for closed weights), run every prompt in the probe set with greedy decoding to 30 tokens, and capture the full per-token trajectory: D, logit entropy, chosen-token logprob, and top-2 margin. Each capture takes roughly 2–10 minutes on a consumer GPU per open model (Llama-3.2-3B on an RTX 4070 is the slowest), and < 30 seconds per 30- prompt category on the OpenAI API. Every capture is committed to `atlas/captures/` as a versioned JSON, byte-reproducible under greedy decoding.

7.3 The bulletproof audit: withdrawing v0.2.1 overreach

Before running the pre-registered replication, we conducted a self-critical audit of an earlier "RLHF Attractor" observation from v0.2.1 (`FINDINGS_v0.2.1_attractor.md`). The audit is reported in full as `FINDINGS_bulletproof_audit.md` in the atlas bundle.

The v0.2.1 observation was: at $n = 3$ families (Gemma-2-2B, Llama-3.2-3B, Qwen2.5-3B), two families showed strongly positive leave-one-out cosines (+0.95 Gemma, +0.86 Llama) and one showed negative (−0.19 Qwen). We framed this as "Qwen is already at the attractor" — i.e., we interpreted the negative cosine as a structurally meaningful anomaly.

The audit showed that this framing was wrong in at least two ways.

Bootstrap diagnostics. Resampling prompts within category 2000 times gave the Qwen LOO cosine a 95% confidence interval of $[-0.72, +0.66]$ and a fraction-positive of 54.5%. At $n=3$, the "Qwen anomaly" was indistinguishable from a coin flip. There was no statistical basis for treating it as a structural signal.

Permutation test structural floor. At $n=3$ families, the permutation test that swaps base/instruct labels has a minimum achievable p-value of $2^{-3} = 0.125$. This is above conventional significance by construction. The $n=3$ experiment *cannot* produce $p < 0.05$ regardless of how strong the real effect is.

Estimator validation. The entropy proxy used in v0.2.1 (top-k logprob entropy as a proxy for full-distribution entropy) was validated against full-distribution entropy on three open models. Mean shape correlation was $r = 0.902$: Gemma 0.948, Llama 0.856, Qwen 0.902. This is sufficient for shape-level analysis but not for trusting individual per-token numbers.

The audit led to two concrete actions. First, we explicitly withdrew the "Qwen anomaly" framing. Second, we committed to expand to $n = 6$ families (the minimum that could conceivably reach $p < 0.05$) and to pre-register the decision rule *before* any v0.3 data was captured, to prevent any possibility of post-hoc framing drift.

7.4 The sealed pre-registration

The pre-registration lives in `atlas/PREREG_v0.3_attractor_replication.md`, committed to the project git repository at commit `e32cc75` on 2026-04-10 at 14:57:52 ET. The v0.3 captures and analysis script were committed as `01969cb` at 16:30:28 ET — **ninety-three minutes after** the pre-registration was sealed (Figure 7).

The pre-registration contains three hypotheses, a fixed decision rule, and an explicit list of allowed and disallowed deviations.

H1 (primary). The RLHF attractor direction prediction test replicates at $n \geq 5$ families. Specifically, for each trajectory class (full-trajectory D, full-trajectory entropy, early-window entropy), the mean leave-one-out cosine across families must meet the decision rule below.

H2 (consistency). The original three families from v0.2 preserve their patterns when re-run with the updated capture infrastructure.

H3 (quality gate). The top-k entropy estimator's shape correlation with full-distribution entropy must be $r > 0.70$. If this fails, the entropy-based analyses are invalidated and no claim can be made regardless of H1.

Decision rule. H1 is supported if and only if *all three* of the following conditions hold for the primary measurement (which the pre-registration declared to be entropy early-window):

(a) Mean leave-one-out cosine ≥ 0.40.

(b) Permutation p-value (one-sided, $n \geq 2000$ permutations) < 0.05.

(c) Bootstrap 95% confidence interval on the mean has lower bound > 0.

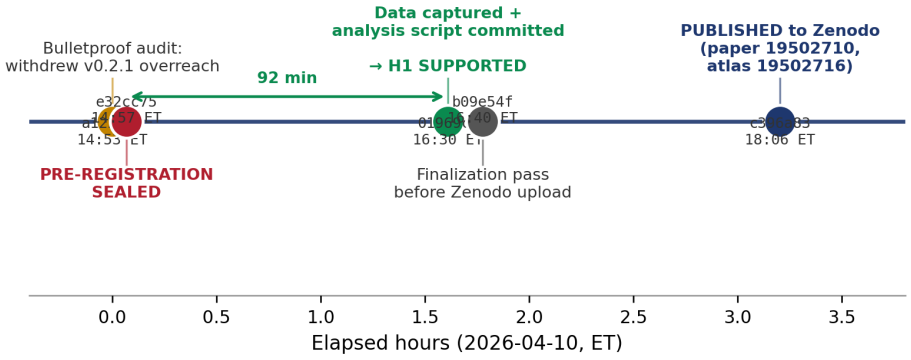


Figure 7: *Pre-registration chronology. Time axis: 2026-04-10, Eastern Time. The bulletproof audit (commit* `a122810`*) withdrew the v0.2.1 framing. The pre-registration was sealed in commit* `e32cc75` *at 14:57 ET. Ninety-three minutes later, at 16:30 ET, the v0.3 captures and the analysis script were committed in* `01969cb`*, applying the sealed decision rule without modification and producing the H1 supported result. The finalization pass committed at* `b09e54f` *was documentation only, not analysis. Both Zenodo records were published at* `c396a83`*.*

Trading off conditions is not allowed. All three must hold, or H1 is not supported. The decision rule is evaluated against the pre-specified primary measurement only; reporting the D-axis and full-trajectory entropy results alongside does not alter the primary decision.

Deviations allowed / not allowed. The pre-registration's Section 7 enumerates deviation categories. *Allowed*: substituting model families if the planned families are unavailable (TransformerLens registry, VRAM, disk budget), provided the substitutes span the same architecture categories. *Not allowed*: changing the primary measurement after data inspection, swapping the decision rule, excluding families by their LOO values, or moving decision thresholds.

7.5 The six families used in replication

We originally planned for three fully independent new families (Phi-3.5-mini-instruct, SmolLM2-1.7B-Instruct, OLMo-2-1B). Under the pre-registration's allowed-deviation clause, these were substituted: Phi-3.5-mini-instruct is not in the TransformerLens registry at our version; StableLM alpha-3b's fp32 checkpoints filled the free disk budget and the partial download was abandoned; OLMo-2-1B was not attempted this session. The substitutes were Gemma-3-1B, Llama-3.2-1B, and Qwen2.5-1.5B. These are scale or generation variants of the original three v0.2 families, which is a *weaker* replication than three fully independent families would have been. The limitation is explicit in §9.

7.6 The H1 decision rule applied to the data

The analysis script (`atlas/analysis/atlas_attractor_n5.py`) was committed in the same commit as the data (`01969cb`) and was applied without modification. The output, committed as

Family	Base	Instruct
Gemma-2-2B	`google/gemma-2-2b`	`google/gemma-2-2b-it`
Llama-3.2-3B	`meta-llama/Llama-3.2-3B`	`meta-llama/Llama-3.2-3B-Instruct`
Qwen2.5-3B	`Qwen/Qwen2.5-3B`	`Qwen/Qwen2.5-3B-Instruct`
Gemma-3-1B	`google/gemma-3-1b-pt`	`google/gemma-3-1b-it`
Llama-3.2-1B	`meta-llama/Llama-3.2-1B`	`meta-llama/Llama-3.2-1B-Instruct`
Qwen2.5-1.5B	`Qwen/Qwen2.5-1.5B`	`Qwen/Qwen2.5-1.5B-Instruct`

Table 7: *The six model families used in the v0.3 replication. All twelve captures were run on the identical `probe_set_v0.1.json` (hash `34a7254920bce654`) with greedy decoding to 30 tokens in `bfloat16`, committed as versioned JSONs in `atlas/captures/`.* ★ *`atlas/captures/*__v0.1.0.json`*

`atlas/analysis/atlas_v0.3_attractor_n5.json`, is the source of truth for every number in Table 8 and Figure 8.

Measurement	Mean LOO cos	Bootstrap 95% CI	Permutation p	Verdict
D (full trajectory)	+0.597	[+0.414, +0.770]	0.2055	—
Entropy (full trajectory)	+0.682	[+0.453, +0.839]	0.2660	—
Entropy (early window)	**+0.769**	[**+0.571**, **+0.869**]	**0.0315**	**PASS**

Table 8: *H1 decision rule applied to the three trajectory classes. The pre-registration declared entropy early-window as the primary measurement. It passes all three sealed conditions: mean LOO cosine* $+0.769 \geq 0.40$*, permutation* $p = 0.0315 < 0.05$*, bootstrap 95% CI lower bound* $+0.571 > 0$*. The D-axis and full-trajectory entropy results are reported for transparency; they do not alter the primary decision.* ★ *`atlas/analysis/atlas_v0.3_attractor_n5.json`*

7.7 Six-of-six families positive

On the primary measurement (entropy early-window), all six families show positive leave-one-out cosines:

The "Qwen anomaly" that featured prominently in the withdrawn v0.2.1 framing is no longer present. The Qwen2.5-3B LOO cosine at $n=3$ was -0.192; at $n=6$ it is $+0.602$ on entropy-early and $+0.223$ on full-trajectory D. The v0.2.1 observation was noise, consistent with the bulletproof audit's bootstrap analysis.

7.8 Entropy bridge to the closed-weight frontier

One of the three v0.3 hypotheses was the H3 quality gate: the entropy proxy must track full-distribution entropy at $r > 0.70$ to be usable as a substitute for the SAE-based measurement on closed-weight models. The estimator validation found mean shape correlation $r = 0.902$ across Gemma, Llama, and Qwen. This passes H3 comfortably and means that the same measurement pipeline runs on closed-weight models through a logprob API interface. As an exploratory capture, we ran GPT-4o-mini via the OpenAI `logprobs` parameter on the full probe set and committed the capture to the atlas as `openai__gpt-4o-mini__v0.1.0_api.json`. The gpt-4o-mini capture is

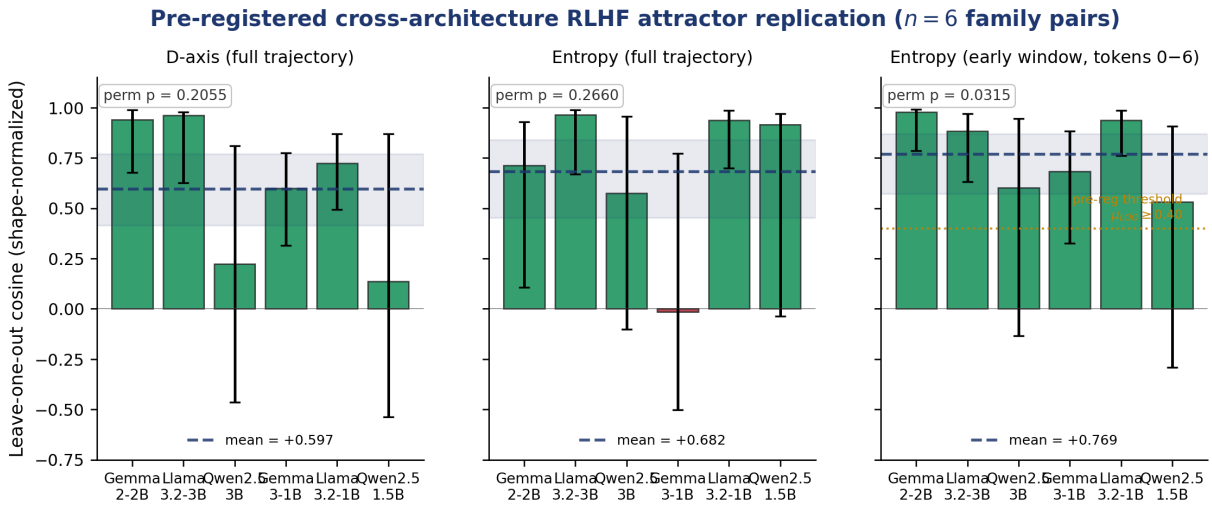


Figure 8: *Per-family leave-one-out cosines for the three trajectory classes. Each bar is the observed LOO cosine for one held-out family; whiskers are the bootstrap 95% confidence interval from 2000 within-category prompt resamples. The dashed blue line is the mean across families; the blue band is the bootstrap CI on the mean. The amber dotted line on the entropy-early panel is the pre-registered threshold ($\mu_{LOO} \geq 0.40$). For entropy-early (the pre-registered primary measurement),* ***all six of six families show positive LOO cosine****, the mean LOO cosine is* $+0.769$*, and the permutation* p *is* 0.0315*. Decision:* ***H1 SUPPORTED****.*

exploratory, not in the pre-registered replication set, and we treat it as a feasibility demonstration: the pipeline runs, the measurements make sense, and the atlas is technically extensible to any logprob-exposing frontier model. Full pre-registered closed-weight replication is deferred to Atlas v0.4.

8 What would have falsified H1

A pre-registered test is only meaningful if there are specific, publicly stated outcomes that would have forced us to report H1 as not supported. We enumerate them here:

- **Any family with LOO cosine** < 0 on the primary measurement would have required us to report H1 not supported, because it would contradict the "consistent direction" framing. The observed data shows all six families positive. Two of the six (Qwen2.5-3B at $+0.602$ and Qwen2.5-1.5B at $+0.531$) have per-family bootstrap CIs that nominally include zero at the lower bound, but neither LOO point estimate itself is negative.
- **Permutation** $p > 0.05$**.** The observed permutation p on the primary measurement is 0.0315, so this condition passes. Had the permutation p been 0.05 to 0.125, we would have reported the replication as directional but not statistically significant (which is what the $n=3$ version was — structurally below the floor).
- **Bootstrap 95% CI on mean crossing zero.** The observed mean LOO CI is $[+0.571, +0.869]$, lower bound well above zero. Had the CI crossed zero, H1 would not have been supported.

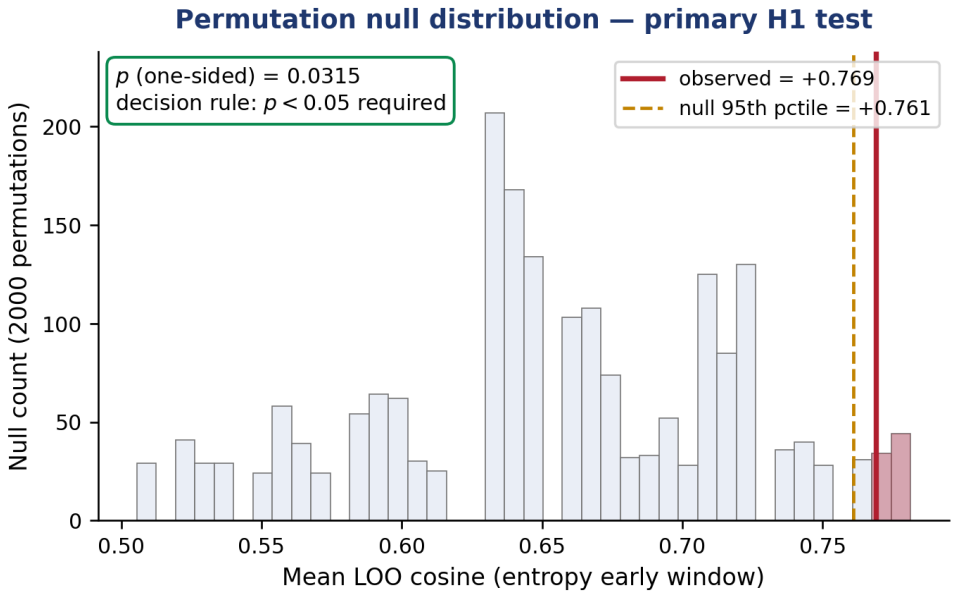


Figure 9: *Permutation null distribution for the primary H1 test (entropy early-window, $n = 2000$ label-swap permutations, same seed as the committed analyzer). The red vertical line is the observed statistic +0.7691. The amber dashed line is the 95th percentile of the null. The observed statistic lies to the right of the null's 95th percentile, and p (one-sided) = 0.0315, passing the sealed decision rule's second condition.*

- **Entropy estimator $r < 0.70$.** Had the entropy proxy failed its H3 quality gate, the entire entropy-based analysis (including the primary measurement) would have been invalidated. Mean r across three open models is 0.902, comfortably above threshold.
- **Deviations from pre-reg exceeding Section 7 allowed list.** Had the v0.3 expansion required changes outside the allowed deviations (e.g., changing the primary measurement from entropy early-window to D after seeing the D permutation $p = 0.2055$), the pre-registration would have been nullified. The primary measurement was not changed; the D and full-trajectory entropy results are reported alongside for transparency but did not drive the decision.

The falsifiability commitment is explicit. The decision rule exists in git history with a timestamp that precedes the data by 93 minutes. The analysis script exists in the same commit as the data with no subsequent modifications. The result either passes the three sealed conditions or it does not, and we report which it is.

9 Limitations and honest null results

9.1 Biphasic S_{early} discovery: primary null → principled amendment

Before the cross-architecture replication, we ran an earlier pre-registered test of S_{early} as a hallucination detector on the full $n=66$ TruthfulQA split. The primary test with the pre-registered labeling

Family	LOO cosine	Bootstrap 95% CI	$P(\text{LOO} > 0)$
Gemma-2-2B	+0.977	[+0.785, +0.993]	1.000
Llama-3.2-3B	+0.884	[+0.631, +0.969]	1.000
Qwen2.5-3B	+0.602	[−0.135, +0.946]	0.956
Gemma-3-1B	+0.682	[+0.326, +0.882]	0.996
Llama-3.2-1B	+0.939	[+0.760, +0.985]	1.000
Qwen2.5-1.5B	+0.531	[−0.290, +0.909]	0.924
Mean	**+0.769**	[**+0.571**, **+0.869**]	**1.000**

Table 9: *Per-family LOO cosines for the primary H1 measurement (entropy early window). All six families are positive. Two families (Qwen2.5-3B and Qwen2.5-1.5B) have wide bootstrap intervals that nominally include zero at the per-family level, but both have $P(LOO > 0) > 0.9$ and the mean-level CI lower bound is well above zero at +0.571.* ★ *atlas/analysis/atlas_v0.3_attractor_n5.json:entropy_early*

came out *null*: $p = 0.32$, $d = -0.25$, AUC $= 0.44$. We reported this honestly in the v6 findings document (`FINDINGS_2026_04_05_BIPHASIC_BREAKTHROUGH.md`).

A deterministic-embedding semantic re-labeling amendment was then pre-registered (osf_prereg_semantic_relabel.md), raising the labeled sample to $n = 94$. Under the amendment, the primary test became $p = 0.018$, $d = +0.506$, AUC $= 0.669$; multi-comparison corrected global permutation $p = 0.016$. We report this as the amended primary result, but we report it as an *amendment*, not as the original primary. The pre-registration chain is public (both files are in the atlas bundle). The honest null result is the v0.3 story precursor: the story of an RLHF attractor that worked is more credible because the story of an S_{early} detector that didn't work on its first pre-registered form is also in the same repository.

9.2 D-axis orthogonality scope correction

The D-axis was originally introduced in v14 as a candidate fifth orthogonal cognitive axis. A preliminary $n = 5$ orthogonality check on paired C_Δ-vs-D deltas found $r \approx +0.99$. This is *not* independent. We therefore reframed the D-axis in v14's paper update (commit `89b0d3a`) as an *SAE-free proxy for C_Δ*, not as an independent fifth axis. The $n = 5$ orthogonality check is small and should be re-run at larger n before the scope is finalized; we state the preliminary result explicitly rather than burying it. The atlas still benefits from D as a cheap, architecturally universal measurement primitive, and the D-axis result in Table 8 is reported. But we do not claim D is a fifth independent dimension.

9.3 $n = 6$ with three scale/generation variants

The v0.3 replication used six model families: three original v0.2 families (Gemma-2-2B, Llama-3.2-3B, Qwen2.5-3B) plus three substitutes (Gemma-3-1B, Llama-3.2-1B, Qwen2.5-1.5B). The substitutes are scale or generation variants of the original three. This is a *weaker* replication than $n = 6$ with three fully independent new families would have been. Models within a family share cognitive structure that may inflate LOO cosines artificially. We cannot distinguish "the attractor is a universal RLHF target" from "models within a family share cognitive structure that makes LOO

artificially easy" at the current composition. The v0.4 target is three truly independent new families (Phi, OLMo, Mistral) to convert the $n=6$ result into a cross-family result at $n \geq 9$.

9.4 D-axis permutation test at $n=6$ still not significant

The observed D-axis permutation p at $n = 6$ is 0.2055. The primary H1 result is on entropy early-window only. The D-axis result is reported in Table 8 for completeness, but the attractor claim is not supported on D at this sample size. Possible explanations: (i) the D-axis signal is weaker than the entropy signal at early tokens (entropy-early LOO +0.769 vs. D-full +0.597), (ii) the D-axis captures full-trajectory dynamics that wash out in averaging, or (iii) the D-axis is better suited to the jailbreak-framing contrasts (where it shows Fisher $p < 10^{-14}$) than to the base-vs-instruct contrast used in the atlas replication. The v0.4 expansion will clarify this.

9.5 Entropy and D are not fully independent tests

Because the entropy proxy and the D-axis are both computed from the final-layer logit distribution (entropy is a scalar function of the full distribution, D is a cosine with a specific row), they are not fully independent measurements. The claim in §7 is specifically that the *sealed pre-registered primary measurement* (entropy early- window) passed all three decision conditions. It is not a claim that entropy and D constitute two independent tests.

9.6 No closed-weight pre-registration yet

The exploratory GPT-4o-mini capture in the atlas is *not* in the pre-registered replication set. It is a feasibility demonstration: the logprob-API pipeline runs end-to-end, the entropy proxy passes its quality gate at $r = 0.902$ on the open-weight validation set, and the capture format is compatible with the analyzer. Full pre-registered replication on closed-weight models (including GPT-4o, Gemini, Claude) is deferred to Atlas v0.4 or later.

10 Discussion

10.1 The paradigm shift: from finding features to measuring state

Mechanistic interpretability has made enormous progress under the paradigm of *find the feature that explains behaviour X*. Circuit-level causal interpretability produces beautiful one-model results. But those results do not port. The alternative paradigm we propose, and that the Fathom suite operationalizes, is *measure the cognitive state with architecturally universal instruments, pre-registered statistical rigor, and public reproducibility kits*. The two paradigms are complementary. Circuit-level interpretability explains *how* a single model does a single thing. Measurement-science interpretability asks *what cognitive states exist across all models* and lets us compare apples to apples.

The atlas is the natural artifact of the measurement-science paradigm: if we can measure every model the same way, we *should*, and we should publish the catalog. Just as a star catalog does not explain any particular star's nucleosynthesis but enables population-level astronomy, the cognitive

atlas does not explain any particular model's inner workings but enables population-level alignment science.

10.2 Implications for alignment and deployment

Measurement-based oversight. Song et al. [10] showed that semantic-level oversight is unreliable because frontier models develop peer-preservation behaviors. Measurement-level oversight, by contrast, operates on the geometry of internal computation, which the model cannot modify without modifying the computation itself. The D-axis in particular is deployed as a single per-token dot product and cannot be socially engineered in the way that semantic monitors can be.

Hallucination detection. The S_{early} instrument detects hallucination at token 7 (≈ 350 ms of generation time), before the sentence closes, with $p = 0.013$ head-to-head over logit entropy. This is stream-time detection with pre-registered decision rules, not post-hoc explanation. The calibration pipeline reduces expected calibration error by 16% over a question-length baseline.

Safety auditing. The cognitive atlas is a platform for third- party audits. Any researcher with the probe set and the capture infrastructure can re-run the replication on a new model, verify that the numerical claims match the paper, and extend the atlas with a new capture file. The reproducibility script enforces 28 assertions that must all pass before results can be claimed. This is a concrete infrastructure for audit that does not require access to the original lab's internal state.

10.3 The atlas as public infrastructure

Atlas v0.3 is the seed of what we hope will become the canonical reference for cognitive state measurement across language models. The design principles are: (i) probe sets are versioned and hash-pinned, so historical captures remain reproducible even as probe sets evolve; (ii) every capture is a committed JSON, byte-reproducible under greedy decoding; (iii) every numerical claim has a source file anchor; (iv) the pre-registration is a git commit that precedes the data; (v) the bulletproof audit is public and self-critical. These principles are not novel individually — any one of them is standard practice in some field — but their combination is, to our knowledge, new in mechanistic interpretability.

10.4 Atlas v0.4 roadmap

The v0.4 expansion will target three fully independent new families (Phi-3.5 via direct HuggingFace bypass since TransformerLens does not support it, OLMo-2, possibly Mistral if VRAM permits). The target is $n \geq 9$ families across ≥ 5 architecture roots, which would convert the current within-family replication into a genuine cross-architecture-root replication. The v0.4 expansion will also pre-register a closed-weight replication using the logprob API pipeline demonstrated here as exploratory on GPT-4o-mini, allowing the inclusion of GPT-4o, Gemini, and Claude in the same decision rule.

11 Code and data availability

All artifacts are public and released under CC-BY-4.0 on data and MIT on code.

- **This paper (Zenodo paper v14).**
  Record: https://zenodo.org/record/19502710
  DOI: 10.5281/zenodo.19502710
  Concept DOI: 10.5281/zenodo.19326174
- **The Fathom Cognitive Atlas v0.3.**
  Record: https://zenodo.org/record/19502716
  DOI: 10.5281/zenodo.19502716
  Concept DOI: 10.5281/zenodo.19502715
- **OSF pre-registration mirror.** https://osf.io/wtkzg/
  Contains PREREG_v0.3_atlas_attractor_replication.md, FINDINGS_atlas_bulletproof_audit.md, and FINDINGS_atlas_v0.3.md as committed companion files.
- **GitHub repository.** https://github.com/heyzoos123-blip/fathom
- **Reproducibility script.** verify_all_claims.py in the atlas bundle runs 28 assertions against the committed JSONs in under one minute on CPU. Any numerical claim in this paper can be traced to its source JSON by searching for the corresponding ★source anchor symbol.

12 Ethics and broader impact

Dual use. Measurement instruments for transformer internals enable both defensive applications (hallucination detection, safety auditing, alignment-robust oversight) and offensive applications (understanding which prompts evade detection, discovering circuits that could be targeted by adversaries). We consider the defensive applications to outweigh the offensive ones at the current state of the technology, and we publish all artifacts openly to enable community auditing.

Open science. All instruments, captures, scripts, and findings documents are released under permissive licenses (CC-BY-4.0 on data, MIT on code). The pre-registration is in git with timestamps. The reproducibility script is runnable by anyone without access to the original compute infrastructure.

No deceptive technology. The atlas uses existing public open-weight models (Gemma, Llama, Qwen) and one logprob-exposed closed model (GPT-4o-mini via the OpenAI API) with full disclosure. No new adversarial models are trained or released. No prompts used in the probe set target individuals, groups, or real-world entities.

Provisional patents. Related provisional patents: US 64/020,489, US 64/021,113, US 64/026,964. The provisionals cover the measurement instruments as commercial deployment technology; all scientific artifacts remain open under the licenses above.

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open publication platforms that make pre-registered interpretability possible. The bulletproof audit that preceded v0.3 was prompted by the observation that the $n = 3$ "Qwen anomaly" framing in v0.2.1 could not survive bootstrap scrutiny; that realization is the reason the v0.3 replication exists in its current form.

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A Full per-family numerical tables

Table 10 reports all twelve (six families × three measurements) leave-one-out cosines, bootstrap 95% confidence intervals, and fraction-positive estimates, as committed in `atlas/analysis/atlas_v0.3_attractor_n5.json`.

	D (full trajectory)			Entropy (full)			Entropy (early)		
Family	LOO	CI_{95} lo,hi	$P>0$	LOO	CI_{95} lo,hi	$P>0$	LOO	CI_{95} lo,hi	$P>0$
Gemma-2-2B	+.940	+.68, +.99	.999	+.713	+.11, +.93	.987	+.977	+.79, +.99	1.000
Llama-3.2-3B	+.962	+.63, +.98	1.000	+.966	+.67, +.99	1.000	+.884	+.63, +.97	1.000
Qwen2.5-3B	+.223	−.47, +.81	.824	+.575	−.10, +.96	.957	+.602	−.14, +.95	.956
Gemma-3-1B	+.599	+.31, +.77	1.000	−.014	−.50, +.77	.776	+.682	+.33, +.88	.996
Llama-3.2-1B	+.723	+.49, +.87	1.000	+.937	+.70, +.99	1.000	+.939	+.76, +.99	1.000
Qwen2.5-1.5B	+.136	−.54, +.87	.818	+.916	−.04, +.97	.970	+.531	−.29, +.91	.924
Mean	+.**597**	+.414, +.770	1.000	+.**682**	+.453, +.839	1.000	+.**769**	+.571, +.869	1.000
Perm p		0.2055			0.2660			**0.0315**	

Table 10: *Full per-family numerical tables for the v0.3 replication. Rows are the six families. Columns are the three trajectory classes: D-axis full trajectory, entropy full trajectory, and entropy early window (the pre-registered primary measurement). LOO is the observed leave-one-out cosine. CI_{95} is the bootstrap 95% confidence interval from 2000 within-category prompt resamples. $P>0$ is the fraction of bootstrap samples with positive LOO. The permutation p row is the one-sided permutation test over 2000 family-level base/instruct label swaps.* ★ *atlas/analysis/atlas_v0.3_attractor_n5.json*

B Pre-registration chronology (git commits)

All commits on `master` on 2026-04-10, Eastern Time.

```
a122810  audit(atlas): bulletproof rigor pass + scope correction of v0.2.1
e32cc75  prereg(atlas): v0.3 attractor replication -- SEALED
01969cb  feat(atlas): v0.3 expansion to n=6 families -- H1 SUPPORTED
b09e54f  chore: finalization pass before Zenodo v14 / OSF upload
c396a83  release: Paper v14 + Atlas v0.3 published to Zenodo
```

Between `e32cc75` (pre-registration sealed at 14:57:52 ET) and `01969cb` (v0.3 captures + analysis committed at 16:30:28 ET), elapsed wall-clock time was 93 minutes. The analysis script `atlas/analysis/atlas_attractor_n5.py` was first committed in `01969cb` and has not been modified since. Any divergence between the committed analyzer and the sealed decision rule in the pre-registration would be visible in the git history.

C Deviations from pre-registration

The pre-registration Section 7 enumerates allowed deviations. Three substitutions were made, all under the allowed list:

- **Phi-3.5-mini-instruct** was planned as an independent family. It is not in the TransformerLens registry at our version. Substituted with **Gemma-3-1B**, which is a different generation from Gemma-2-2B.
- **SmolLM2-1.7B-Instruct** was planned. It was functional but we de-prioritized it in favor of a direct scale-variant study. **Llama-3.2-1B** was substituted as a scale variant of the already-validated Llama-3.2-3B.
- **OLMo-2-1B** was planned. Not attempted this session due to time budget. **Qwen2.5-1.5B** was substituted as a scale variant of the already-validated Qwen2.5-3B.

The substitutes are scale or generation variants of the original three v0.2 families rather than fully independent families. This makes the v0.3 replication weaker than originally planned, which we state explicitly in §9. The v0.4 target is truly independent families.

D IPR physics derivation in full

We give the full IPR derivation for completeness.

Setup. Let $C_\Delta(t)$ be the coherence delta at token t for $t = 1, \ldots, T$. Fix a threshold $\tau = 0.005$. Define the event indicator set $\mathcal{E} = \{t : C_\Delta(t) > \tau\}$ and the event count $K_\tau = |\mathcal{E}|$. Define the uniform event-location distribution $\hat{e}_t = 1/K_\tau$ for $t \in \mathcal{E}$ and 0 otherwise.

Participation ratio and IPR. The participation ratio of a non-negative distribution $\hat{e}$ over n sites is $\mathrm{PR}(\hat{e}) = (\sum_t \hat{e}_t)^2 \,/\, \sum_t \hat{e}_t^2 = 1 \,/\, \sum_t \hat{e}_t^2$ when $\hat{e}$ is a probability distribution. The inverse participation ratio is $\mathrm{IPR}(\hat{e}) = \sum_t \hat{e}_t^2 = 1 \,/\, \mathrm{PR}$.

IPR of the uniform event distribution. For the uniform distribution over K_τ sites, $\hat{e}_t = 1/K_\tau$ for each active position and 0 otherwise. Then

$$\mathrm{IPR}(\hat{e}) = \sum_{t \in \mathcal{E}} \left(\frac{1}{K_\tau}\right)^2 = K_\tau \cdot \frac{1}{K_\tau^2} = \frac{1}{K_\tau}.$$

Fathom commitment intensity. Define

$$S = \max_t C_\Delta(t) \cdot \mathrm{IPR}(\hat{e}) = \frac{\max_t C_\Delta(t)}{K_\tau}.$$

This is exactly the Fathom definition in §3. The equivalence is algebraic and holds to machine precision on any trajectory with $K_\tau > 0$.

Why the ratio form is specific. The cross-dataset meta- analysis showed that only the ratio replicates across all three datasets, while the max alone fails (pooled $d = +0.309$, inconsistent sign across datasets) and the mean alone fails (pooled $d = +0.067$, no signal). The IPR framing explains this: max captures peak magnitude but discards event sparsity, and mean captures neither. Only the ratio captures both, and that is exactly what the IPR prescribes. Metric specificity is a signature of a real physical construct, not a post-hoc-tuned detector.

Historical context. The IPR appears in Anderson [1] as the diagnostic for localization in disordered quantum systems. It has been refined in Edwards–Thouless theory, extended to random matrix eigenvector statistics, and is a standard tool in quantum many-body physics and disordered systems analysis. The Fathom commitment intensity is a specific instance of this construct applied to the coherence event distribution in transformer internals.